

Topical Computer Science: Popular Papers

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Purpose

Our project aims to analyze the DBLP database to evaluate the popularity of different topics within the field as well as to find correlations in their importance over time. By identifying topics and trends that have grown, declined, or remained consistent in visibility, the project aims to find insight into how the overall discipline has evolved over time. Different methodologies were used to establish the basis of measurement and comparison, which can be seen listed below.

Methodology & Process

Data Preprocessing

The first stage consisted of preprocessing the large dataset to better understand its overall structure, quality, and characteristics. We merged the separate JSON files into a parquet file.. Missing data was filled by using median and zero-score imputation. Additionally, we applied feature engineering to create new distinct features, such as graph embeddings from the id←references relationship, TF-IDF vectorization of the combined text in each paper, and converting the rest of our categorical columns into numerical features. Lastly, normalization was applied to ensure similar scales across the board.

At the end, our dataset comprised 2 tables. Our first table was a modified version of our original table, which added the number of authors as a new feature, and also combined all of the text in several columns. The text_combined column was formulated by the following:

$$(title * 2 + abstract + venue + " ".join(authors))$$

We doubled the title column text to highlight its importance alongside the rest of the text.

We stored the results of the TF-IDF vectorization into a second table, which was about 10,000 individual features in a sparse matrix. We initially tried to combine the TF-IDF columns

with the rest of the table using dimensionality reduction (with the help of *scikit's TruncatedSVD*), but we ran into problems with interpretability and performance (which we cover in *Challenges*).

Exploratory Data Analysis

In this stage, exploratory data analysis was conducted to find patterns through statistical analysis and visualization. Correlations of features were identified and plotted in order to find any variance to help with future stages. Outlier detection was also included in this stage in order to improve any noise or irrelevant parts of data which could be removed. The main feature being compared was the publication year, as this was the core concept being explored for the project. Lastly, this stage can provide an insight into overarching trends that can be explored further using classification or clustering methods.

Topic Clustering/Semi-Supervised Learning

We then focused on class label creation, where the goal was to find underlying paper topics by finding natural groupings via clustering. By using the results of the TD-IDF vectorization of the combined text, we ran Non-Negative Matrix Factorization (NMF) to extract distinct 25 topics from the dataset. After that, we looked at the top 20 words of each topic and created a label for each. We grabbed the top 1000 most representative samples from each topic cluster by sorting them by the NMF output weights. Using those labels, we trained a logistic regression model using Stochastic Gradient Descent, and then used that model to predict the labels for the rest of the papers.

Topic Classification

For classification, each paper was first assigned a topic label (one of 25 research categories) by a previously run clustering pipeline, producing a label column called `predicted_topic` in `df_umap_labeled.parquet`. The classification task is to make a model learn to

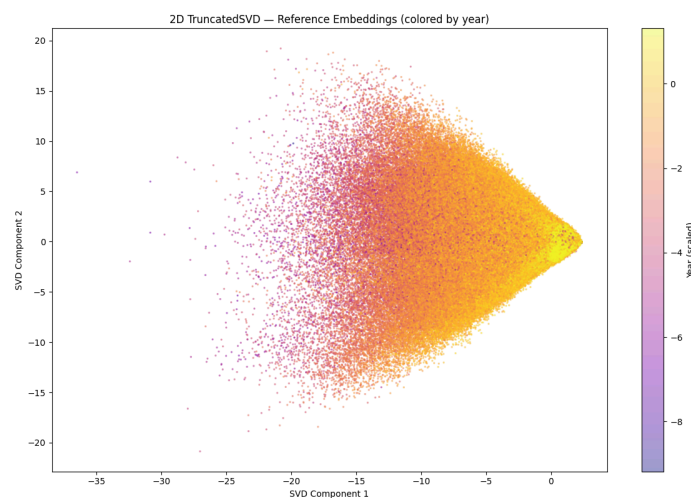
reproduce and generalize the topic assignments from each paper's title and abstract text. Three models were used to evaluate this data set in terms of classification. The first one was Multinomial Naive Bayes (MNB) which is well-suited to sparse TF-IDF features. The second one was building a Decision Tree for interpretability. The path from root to leaf directly reveals which vocabulary terms are most discriminative. Four depth and min-split configurations were tested. The final one used was the Random Forest ensemble of 50 to 150 trees. Due to memory limits, the RF sweep trained on a 50,000-paper subsample with `max_samples = 0.5` per tree. All twelve configurations were evaluated on the same 60,000-paper test set using Accuracy and Macro-averaged F1. Macro-F1 is the primary metric because it treats all 25 topics equally, penalizing models that ignore small classes.

Exploratory Data Analysis

Using the new class labels, we created several visualizations centered around the amount of citations each topic had. Some of the generated graphs include the top 5 topics each year by number of citations and the amount of citations aggregated by topic over the years.

Results

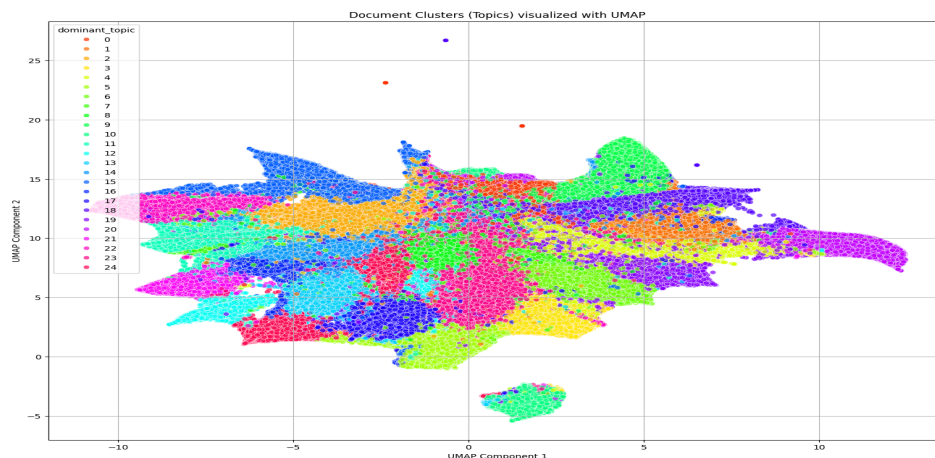
Preprocessed EDA



The 2D SVD shows the reference embeddings which represent the relationships between the citations of papers. The color coding represents the year to get an insight into the trends. One of the things that is seen is the gradient of coloring which shows that the nature of references and citations for papers has changed a lot over time. Each group of year shows up clustered together indicating the research in topics from one period of time is similar to others of the same time but different than other time periods. The shape of the overall graph also shows that papers are closer grouped together showing more cross referencing between papers compared to the further spread from older papers. This shows that older papers could be establishing frameworks for topics while newer papers are possibly similar in their foundations. Overall, this graph shows the existence of a connection and trend that exists between the references in the papers across the years.

Additionally, we looked at some specific paper titles + abstracts. In particular, we noticed that abstracts varied greatly in length, from some being a thorough explanation of the paper's materials, vs. others just including copyright information. Usually, the papers we noticed had more thorough explanations and were more focused on numerical analysis studies, which explains some of our clustering results later in our processing.

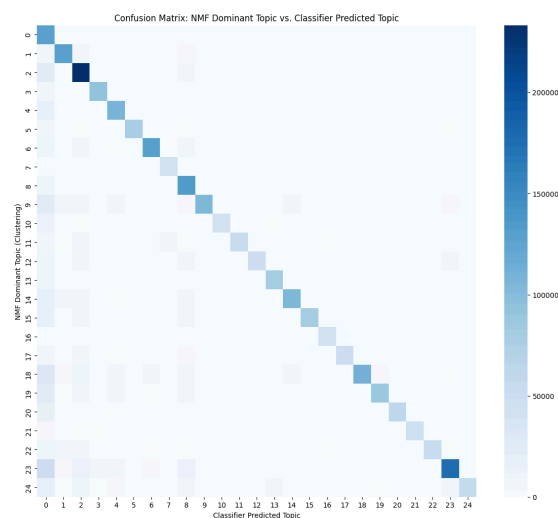
Clustering



Using our TD-IDF $\rightarrow$ NMF pipeline, we were able to successfully extract 25 distinct topics from our document clusters, which is represented in the above graph with a UMAP dimensionality reduction. We can clearly visualize 25 individual clusters with slight overlap (which is expected as many papers don't necessarily consist of just a single topic). After looking at the top words of each cluster, we came up with these class labels:

0. Numerical Methods 1. Networking 2. Information Systems, Business 3. Digital Image Processing 4. Algorithms & Data Structures 5. Machine Learning 6. Data Science 7. Search Engine, Semantics 8. Statistical Modeling 9. Signal Processing 10. Fuzzy logic 11. Cloud Infrastructure 12. Speech/Face Recognition 13. Control Systems 14. Power 15. Software Engineering Principles 16. Mobile 17. Agentic AI 18. Performance 19. Scheduling 20. Graphing 21. Video Processing 22. CyberSec 23. Classification 24. Robotics

After training the logistic regression classifier on our 'confident' labels, we then achieved labeling on the rest of the dataset. Overall, there was a 75% agreement between the classifier and clustering model on the results of labeling, showing a significant difference in the most 'confident' data points vs. the more general clustering efforts.



Clustering vs. Classifier Predicted Topics

Taking a closer look at individual data points, we also noticed that some data points that were better suited to DIP and A&DS were assigned to Numerical Methods. Overall, Numerical Methods was the most dominant class label. As to why, it's most likely that most of the documents utilized heavy mathematical analysis in their research, and therefore a lot of mathematical language. As a result, the NMP model grouped a large portion of documents by their mathematical language under Numerical Methods.

Classification

Model Configuration	Accuracy	Macro F1	Macro Prec.	Macro Rec.	Time
Decision Tree depth=None min_split=20	0.7406	0.7641	0.7758	0.7535	733.88 s
Decision Tree depth=30 min_split=5	0.6735	0.7237	0.8079	0.6733	225.10 s
Decision Tree depth=20 min_split=10	0.6031	0.6357	0.7863	0.5748	115.93 s
Random Forest n=150 depth=24 leaf=4	0.6167	0.6314	0.8373	0.5453	3.32 s
Random Forest n=100 depth=20 leaf=4	0.5581	0.5535	0.8261	0.4724	1.78 s
MNB alpha=0.1 (best NB)	0.5564	0.5426	0.6037	0.5228	0.37 s
MNB alpha=0.5	0.5582	0.5421	0.6240	0.5152	0.36 s
MNB alpha=0.01	0.5543	0.5397	0.6006	0.5191	0.48 s
MNB alpha=1.0	0.5541	0.5308	0.6438	0.4955	0.35 s
Random Forest n=100 depth=16 leaf=2	0.4850	0.4617	0.8157	0.3844	1.44 s
Decision Tree depth=10 min_split=20	0.4583	0.3588	0.7359	0.3366	74.44 s
Random Forest n=50 depth=12 leaf=2	0.3806	0.3007	0.7812	0.2569	0.53 s

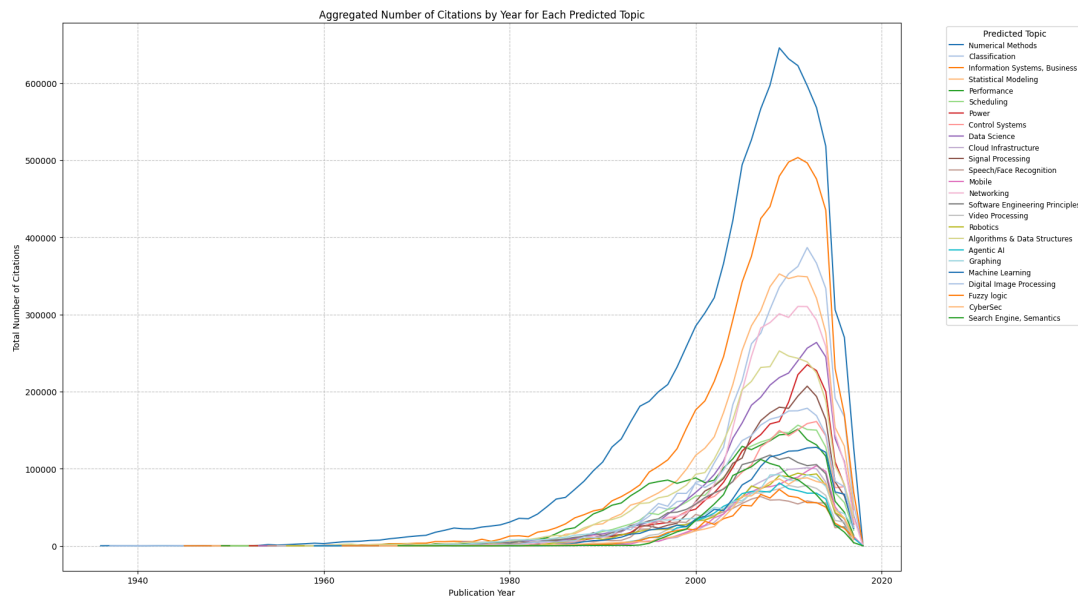
Best model highlighted in green: an unconstrained Decision Tree (depth=None, min_split=20) reached Macro-F1 = 0.7641 on the 60,000-paper test set.

The master comparison table (figure to the left) presents all twelve model configurations evaluated on the same 60,000 paper test set ranked by Macro F1 from best (top) to worst (bottom). The decision tree with no depth cap and a minimum slope of 20 produced the best Macro F1 result of 0.7641 and 74% accuracy. The fact that the Macro F1 for that was higher than the accuracy proves that it

performed consistently across both large and small topics. The four Naive Bayes configurations are clustered together in the middle of the table and have extremely similar scores despite the four different alpha values tested. This similarity shows that the NB model was actually limiting itself since it assumes every word is unrelated to every other word. This caused it to fail to grasp interactions between overlapping topics. The results from Random forest appear all over the table, but this can be attributed more so to the fact that the RF model was trained on only 50,000 rows compared to the Decision Tree that was able to be trained on 240,000 rows. This discrepancy was caused due to memory constraints but an argument could be made that the gap

between RF and Decision Trees scores would be much closer if they had the same number of rows trained on. Overall, this table shows that TF-IDF features from only titles and abstract are strong enough to achieve over 74% accuracy across 25 research topics. The constraints on models also greatly affected results as the Decision Tree model was able to gain almost 40 Macro F1 points from just adjusting the depth.

Postprocessed EDA



After we could label our documents, we wanted to assess popularity via the number of citations. By aggregating the number of citations by topic, we visualized the citations per topic over the publication years. We can see that Numerical Methods, alongside Classification and Information Systems, made up the top labels of the number of citations. We can also see some interesting trends, such as the number of citations having a sharp increase past 1990, likely due to the increase of computing power + the proliferation of the Internet. Relatively new topics, such as Agentic AI and Cloud Infra, were limited by their age.

Ideally, we wanted to capture topics that were less popular, and examine as to why. However, the nature of clustering made it so outliers were either removed or forced into other

clusters. We were planning on using several clustering models (such as DBSCAN, which could identify anomalies), but the large size of the dataset made it impossible.

Analysis

After using a couple of different models, we were able to successfully obtain labels for most of the data. The data clearly showed a delineation based on our NLP steps, and several manual inspections made it clear that the created labels matched most of the titles/abstracts.

However, on closer inspection, it was difficult to analyze our results for the purpose of identifying the least popular topics. As explained before, the nature of clustering and dimensionality reduction flattened our results (ultimately to 25 features), so it was difficult to extract less popular topics. A lot of the extracted patterns were also due more to how research is ultimately iterative: As more research is done, there are more papers in that topic, and therefore more citations + references.

Despite these limitations, the project demonstrated that NLP preprocessing combined with clustering + semi-supervised learning can effectively uncover thematic structures within large collections of academic papers. Future work could focus on methods better suited for anomaly and minority-topic detection, such as DBSCAN or hierarchical clustering, as well as experimenting with richer embeddings and less aggressive dimensionality reduction techniques. Overall, the results suggest that machine learning approaches can provide meaningful insight into research trends while also highlighting the challenges of representing less common topics within large-scale academic datasets.

Challenges

Throughout the implementation of this project, we faced a few challenges. One of the primary challenges that we encountered was working with a large dataset. The 3M rows + high dimensionality of text modeling made computation either very slow or near impossible, which led to having to separate the text and other features into separate tables. We also tried to cluster our dataset using all of the features, but we found that dimensionality reduction methods such as TruncatedSVDs and PCAs could not maintain the dataset's variability (which can be seen in the ClusteringFail). Other computational issues we faced were runtime issues with Google Colab; forcing us to use the platform's high-RAM capacity. Lastly, finding a way to combine four separate methods into a combined structure added complexity which required coordination between all team members being able to do their part accurately and with consistency throughout. All team members were responsible for one method, Aqeel was responsible for the preprocessed EDA, Ricardo was responsible for the clustering, and Santiago was responsible for the classification. All team members were also responsible for writing about their respective method in the report. Work was also done across different methods by everyone, to ensure proper cohesion for working with the data and proper flow in the report.